

Credit Card Approval Prediction

Phase 3: Project Design Phase

SmartBridge / SkillWallet Virtual Internship Program

Team Member	Role
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1. System Architecture

The system is organised into a data layer, a preprocessing/model-training layer, and a Flask serving layer. Raw applicant and credit history CSVs are cleaned, merged, and feature-engineered; four classifiers are trained and compared; the best model is serialized with Pickle and loaded by the Flask app to serve real-time predictions.

Credit Card Approval Prediction — Detailed Architecture (SmartBridge Version)

Two-source dataset -> merge & feature engineer -> train 4 models -> Flask app

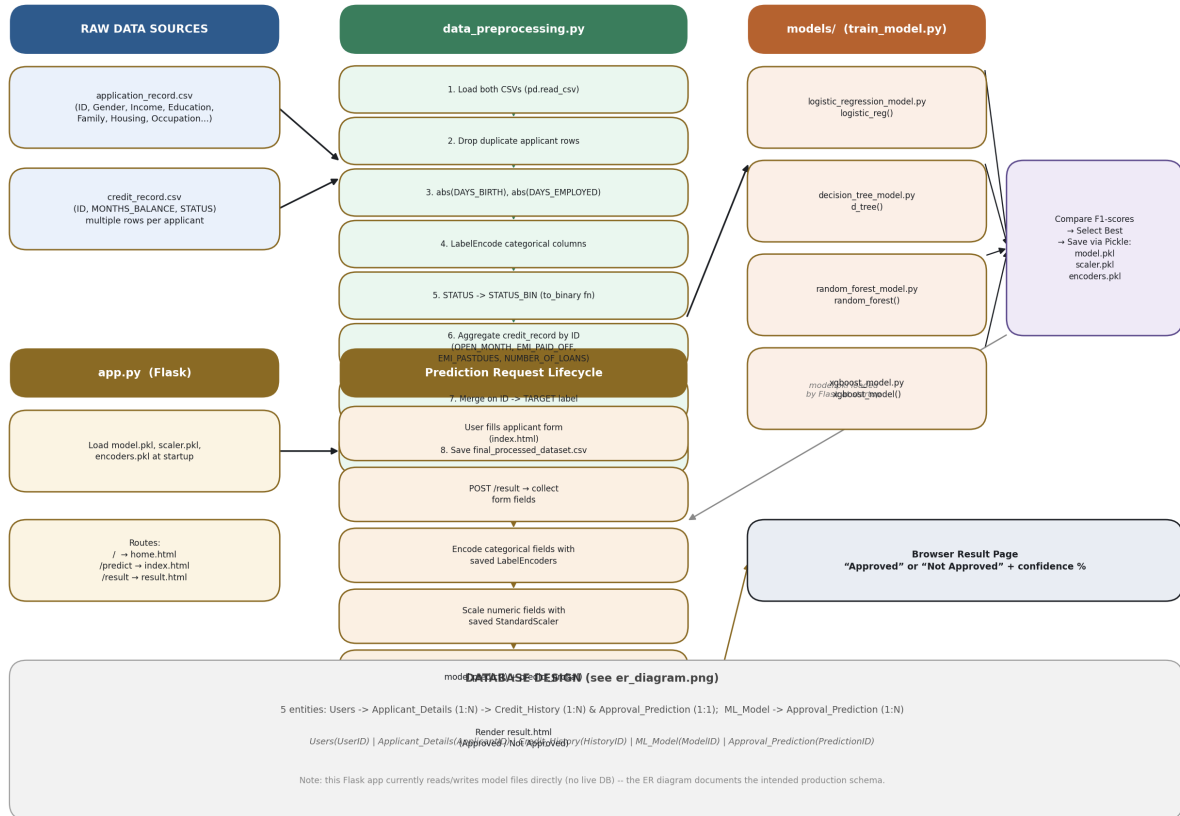


Figure 1 — Detailed system architecture and data flow.

2. Database Design — Entity Relationship Diagram

Five entities model the production database design: Users, Applicant_Details, Credit_History, ML_Model, and Approval_Prediction.

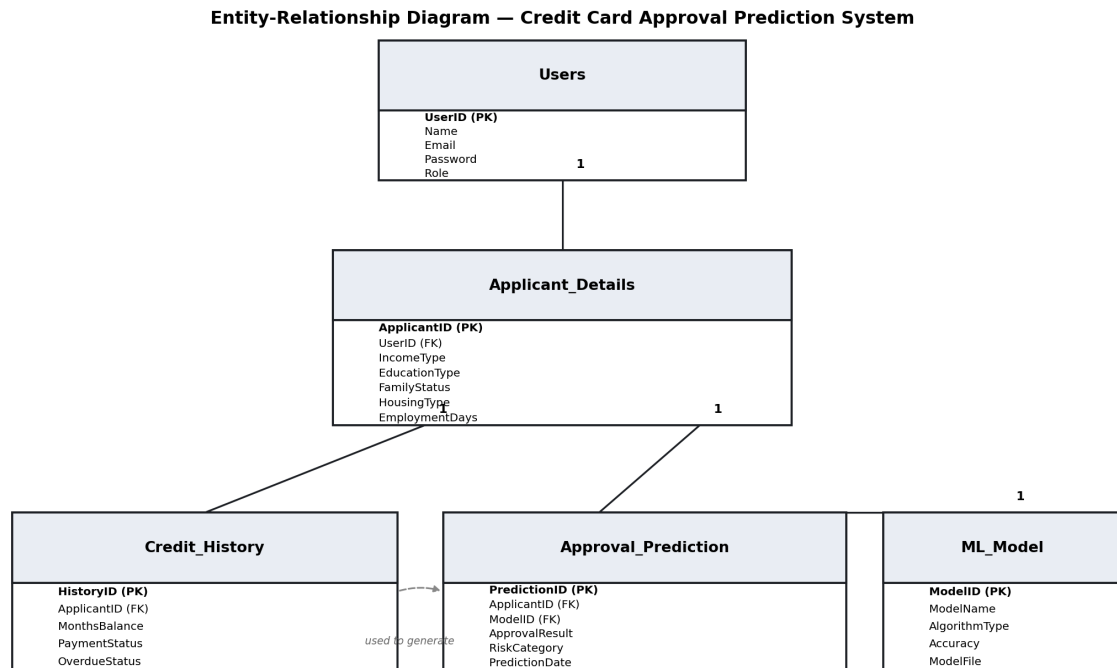


Figure 2 — Entity-Relationship diagram.

3. Data Flow Diagram (Level 0)

- **Input:** applicant fills the web form → Flask collects raw form values.
- **Process 1 — Encode:** categorical fields transformed with saved LabelEncoders.
- **Process 2 — Scale:** numeric fields transformed with the saved StandardScaler.
- **Process 3 — Predict:** the feature vector is passed to the trained model.
- **Output:** Approved/Rejected decision + confidence score rendered back to the user.

4. UI Design

A single-page Flask application (index.html) presents a form for applicant details, grouped into two-column rows for compactness: Gender/Own Car, Own Realty/Family Status, Income Type/Education, Housing Type/Annual Income, Days Birth/Days Employed, Family Members/Number of Loans, EMI Paid Off/EMI of Pastdues. Submitting the form re-renders the same page with a colour-coded result banner — green for Approved, red for Rejected — along with the model's confidence percentage and the name of the deployed model.

5. Technology Stack

Layer	Technology
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn (Logistic Regression, Decision Tree, Random Forest), XGBoost
Model Persistence	Pickle
Web Framework	Flask (Jinja2 templates, HTML, CSS)
Cloud Deployment (optional)	IBM Watson Machine Learning, IBM Cloud Object Storage
Development Environment	Anaconda Navigator, PyCharm, Jupyter Notebook